Category	Method / Command	Description
Connection	sqlite3.connect('file.db')	Creates or opens a physical database file on the disk.
Connection	sqlite3.connect(':memory:')	Creates a temporary database entirely in RAM.
Cursor	db.cursor()	Creates a cursor object used to execute SQL commands.
Execution	cursor.execute(sql, [params])	Executes a single SQL statement. Use ? for variables.
Execution	cursor.executemany(sql, list)	Executes the same SQL command for every item in a list of tuples.
Retrieval	cursor.fetchone()	Retrieves the next single row from a query result.
Retrieval	cursor.fetchall()	Retrieves all remaining rows from a query result as a list.
Retrieval	cursor.lastrowid	Returns the ID of the most recently inserted row.
Transaction	db.commit()	Saves all changes made during the current transaction to the database.
Transaction	db.rollback()	Undoes all changes made since the last commit.
Cleanup	db.close()	Closes the database connection; uncommitted changes will be lost.
Utility	db.row_factory = sqlite3.Row	Configures the cursor to allow column access by name (e.g., row['name']).

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SQLite3 Cheat Sheet

SQLite is a public domain C-language library implementing a small, fast, self-contained, reliabile, and full-featured, SQL database engine.

Manipulating data

Create database	`> .open example.db;`
Create table and define fields	`> CREATE TABLE IF NOT EXISTS mytable (` `→ foo TEXT NOT NULL);`
View tables in database	`> .tables`
Insert data into a table	`> INSERT INTO mytable (foo)` `→ VALUES ('aaa'), ('bbb'),('ccc');`
View table schema	`> .schema mytable`
Add a new column to **mytable**	`> ALTER TABLE mytable ADD bar INTEGER;`
Update data in a table	`> UPDATE mytable SET bar=123` `→ WHERE foo='aaa';`

Joins

Display an inner join	`> SELECT * FROM mytable`
	`→ INNER JOIN othertable`
	`→ ON mytable.rowid=othertable.foo;`

Display a left join	`> SELECT * FROM mytable LEFT JOIN` `→ ON mytable.id=othertable.foo;`
Display a cross join	`> SELECT * FROM mytable` `→ CROSS JOIN othertable;`

Data types

TEXT	Text data
INTEGER	Whole number
REAL	Fleating point number
BLOB	Binary data
NULL	Null value

Some SQLite functions

`abs()`	Absolute value
`max() min()`	Maximum and minimum values
`upper() lower()`	Convert case of string
`length()`	Length of string
`random()`	(Pseudo) random integer

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Select

Display all data	`> SELECT * FROM mytable;`
Display data of the third row	`> SELECT * FROM mytable` `→ WHERE rowid=3;`
Display **foo** and **bar** columns	`> SELECT foo,bar FROM mytable;`
Display first 10 results	`> SELECT * FROM mytable LIMIT 10;`
Sort by column **foo**	`> SELECT * FROM mytable ORDER BY foo;`

Views

A view is a virtual table providing a template for displaying the results of a specific query.

Create a new view	`> CREATE VIEW myview AS`
	`→ SELECT foo FROM mytable`
	`→ WHERE example > 10;`

Show existing views	`> .tables`
Display data with a view	`> SELECT * FROM myview;`
Delete (*drop*) a view	`> DROP VIEW myview;`

Column constraints

Set default text for a field	`DEFAULT 'default text'`
Enforce unique value	`UNIQUE`

Designate a column as a unique identifier	`PRIMARY KEY`
	`> CREATE TABLE mytable`
	`→ (Id INTEGER PRIMARY KEY);`

Pointer to a primary key of a different table	`FOREIGN KEY`

Impose a condition for validation	`CHECK`
	`> CREATE TABLE mytable`
	`→ (CHECK(condition>0), bar TEXT);`

Prevent NULL values	`NOT NULL`